EXP. NO: 07
DATE: 10/09/26.

Develop a Simple Calculator using XML-RPC.

Aim:

To develop a simple calculator using XML-RPC in python to perform basic arithmetic operations between a client and a server.

Procedure:

1. Create folder named XMLRPC_calculator
2. Create two files: server.py, client.py.
3. Write XML-RPC server.py.
4. Write server using python server.py.
5. Write client program in client.py.
6. Open another cmd and run python client.py.
7. Enter 2 numbers select addition, sub, multiplication or division.

Result:

Thus a simple calculator was successfully developed using XML-RPC in python and operations were performed b/w server and client.

server.py - C:\Users\HDC0222020\Desktop\XMLRPC_Calculator\server.py (3.14.5)

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```
from xmlrpc.server import SimpleXMLRPCServer

server = SimpleXMLRPCServer(("localhost", 8000))
print("XML-RPC Calculator Server Running on Port 8000...")

def add(a, b):
    return a + b

def subtract(a, b):
    return a - b

def multiply(a, b):
    return a * b

def divide(a, b):
    if b == 0:
        return "Division by zero is not allowed."
    return a / b

server.register_function(add, "add")
server.register_function(subtract, "subtract")
server.register_function(multiply, "multiply")
server.register_function(divide, "divide")

server.serve_forever()
```

client.py - C:\Users\HDC0222020\Desktop\XMLRPC_Calculator\client.py (3.14.6)

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```
import xmlrpc.client

server = xmlrpc.client.ServerProxy("http://localhost:8000")

a = int(input("Enter First Number: "))
b = int(input("Enter Second Number: "))

print("\nSelect Operation")
print("1. Addition")
print("2. Subtraction")
print("3. Multiplication")
print("4. Division")

choice = input("Enter your choice (1-4): ")

if choice == '1':
    print("Result = ", server.add(a, b))
elif choice == '2':
    print("Result = ", server.subtract(a, b))
elif choice == '3':
    print("Result = ", server.multiply(a, b))
elif choice == '4':
    print("Result = ", server.divide(a, b))
else:
    print("Invalid Choice")
```

C:\Windows\system32\cmd.exe - python server.py

```
C:\Users\HDC0222020\Desktop\XMLRPC_Calculator>python server.py
XML-RPC Calculator Server Running on Port 8000...
127.0.0.1 - - [10/Sep/2026 08:31:32] "POST /RPC2 HTTP/1.1" 200 -
```

C:\Windows\system32\cmd.exe - python client.py

```
C:\Users\HDC0222020\Desktop\XMLRPC_Calculator>python client.py
Enter First Number: 20
Enter Second Number: 10

Select Operation
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice (1-4): 1
Result =  30
C:\Users\HDC0222020\Desktop\XMLRPC_Calculator>
```